

2.15 (C) Consider the probability space $([0,1], \mathcal{B}([0,1]), m)$, where m is the Lebesgue measure

(c) Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a cdf. For $0 < x < 1$, let

$$F_1^{-1}(x) = \inf \{y: y \in \mathbb{R}, F(y) \geq x\}$$

$$F_2^{-1}(x) = \sup \{y: y \in \mathbb{R}, F(y) \leq x\}$$

Let Z_i be the random variable defined by

$$Z_i = F_i^{-1}(x) \quad 0 < x < 1 \quad i=1,2.$$

(i) Find the cdf of Z_i , $i=1,2$.

(ii) Show also that $F_1^{-1}(\cdot)$ is left continuous and $F_2^{-1}(\cdot)$ is right continuous.

(i) the cdf of Z_i is defined as $F_{Z_i}(z) = F_i(1 - m(Z_i \leq z))$

For any $x \in (0,1)$ suppose $F(t) \geq x$

$$\Rightarrow t \in \{y: y \in \mathbb{R}, F(y) \geq x\} \Rightarrow t \geq F_1^{-1}(x) = Z_1$$

Also, suppose $t \geq F_1^{-1}(x) \quad \forall x \in (0,1)$

$F_1^{-1}(x)$ is the inf of $\{y: y \in \mathbb{R}, F(y) \geq x\}$

$\exists \{y_n\}$ in $\{y: y \in \mathbb{R}, F(y) \geq x\}$ s.t.

$y_n = F_1^{-1}(x)$ for all n , and $\lim_{n \rightarrow \infty} y_n = F_1^{-1}(x)$

By right-continuity of CDF F , $F(t) \geq F(F_1^{-1}(x)) = \lim_{n \rightarrow \infty} F(y_n) \geq x$

So $t \geq F_1^{-1}(x) \Leftrightarrow F(t) \geq x$

$$\Rightarrow P(Z_1 \leq t) = m(\{x: F_1^{-1}(x) \leq t\}) = m(\{x: x \leq F(t)\}) = F(t)$$

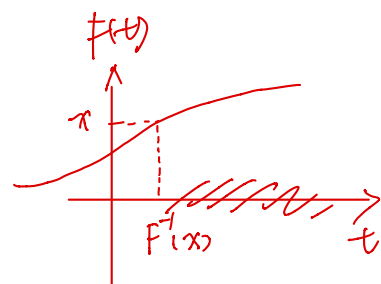
Similarly, for every $x \in (0,1)$ suppose $F(t) \leq x$

$\Rightarrow \forall y \in \{y: y \in \mathbb{R}, F(y) \leq x\} \quad t \geq y$ (F is non-decreasing)

$$\Rightarrow t \geq \sup \{y: y \in \mathbb{R}, F(y) \leq x\} = F_2^{-1}(x) = Z_2$$

Also, suppose $t \geq F_2^{-1}(x) \quad \forall x \in (0,1)$

$F_2^{-1}(x)$ is the sup of $\{y: y \in \mathbb{R}, F(y) \leq x\}$



$$\Rightarrow F(t) \geq x$$

$$\Rightarrow F(t) \geq x \Leftrightarrow t \geq F_2^{-1}(x) = z_2$$

$$P(Z_2 \leq z) = P(F_2^{-1}(x) \leq z) = m(\{F_2^{-1}(x) \leq z\}) = m(\{x \leq F(t)\}) \\ = F(z)$$

(ii) $\forall \varepsilon > 0, \exists \delta_\varepsilon > 0$ s.t. $F(F_1^{-1}(x) - \varepsilon) < x - \delta_\varepsilon$

since $F(t) \geq x \Leftrightarrow F_1^{-1}(x) \leq t$

$$\Rightarrow F_1^{-1}(x - \delta_\varepsilon) > F_1^{-1}(x) - \varepsilon$$

$$\Rightarrow \forall \varepsilon > 0, \exists \delta_\varepsilon > 0, \text{ s.t. } x' \in (x - \delta_\varepsilon, x)$$

$$\Rightarrow F_1^{-1}(x) - F_1^{-1}(x') \leq F_1^{-1}(x) - F_1^{-1}(x - \delta_\varepsilon) < \varepsilon$$

$$\Rightarrow F_1^{-1} \text{ is left-continuous.}$$

similarly, $\forall \varepsilon > 0, \exists \delta_\varepsilon > 0$ s.t. $F(F_2^{-1}(x) + \varepsilon) > x + \delta_\varepsilon$

$$\Rightarrow F_2^{-1}(x + \delta_\varepsilon) \leq F_2^{-1}(x) + \varepsilon$$

$$\Rightarrow F_2^{-1} \text{ is right-continuous.}$$

2.20. Apply Corollary 2.3.5 to show that for any collection $\{a_{ij} : i, j \in \mathcal{N}\}$ of nonnegative numbers

$$\sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right) = \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right)$$

Let $f_i(x) = \sum_{j=1}^{\infty} a_{ij} I_{[j, j+1)}(x)$, $i=1, 2, \dots$ on a measure space $(\mathcal{B}, \mathcal{F}, \mu)$

$$\Rightarrow f_i \text{ is a nonnegative measurable function for every } i.$$

By corollary 2.3.5. $\int_{\mathcal{R}} \left(\sum_{i=1}^{\infty} f_i d\mu \right) = \sum_{i=1}^{\infty} \int_{\mathcal{R}} f_i d\mu.$

$$\int_{\mathcal{R}} f_i d\mu = \int_{\mathcal{R}} \sum_{j=1}^{\infty} a_{ij} I_{[j, j+1)}(x) d\mu = \sum_{j=1}^{\infty} \int_{\mathcal{R}} a_{ij} I_{[j, j+1)} d\mu = \sum_{j=1}^{\infty} a_{ij}$$

$$\Rightarrow \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right) = \int_{\mathcal{R}} \left(\sum_{i=1}^{\infty} f_i d\mu \right)$$

And. $\int_{\mathcal{R}} \sum_{i=1}^{\infty} f_i d\mu = \int_{\mathcal{R}} \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right) I_{[j, j+1)}(x) d\mu = \sum_{j=1}^{\infty} \int_{\mathcal{R}} \left(\sum_{i=1}^{\infty} a_{ij} \right) I_{[j, j+1)}(x) d\mu$
 $= \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right)$ By corollary 2.3.5 again.

$$\Rightarrow \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right) = \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right)$$

2.23. If $f(x) = \mathbb{I}_{\mathbb{Q}}(x)$ where $\mathbb{Q}_1 = \mathbb{Q} \cap [0,1]$, $\mathbb{Q}$ being the set of all rationals. then, show that for any partition P , $U(P, f) = 1$ and $L(P, f) = 0$.

$f(x) = 1$ if x is an rational number in $[0,1]$ and 0 otherwise.

For any finite partition $P = \{x_1, x_2, \dots, x_n\}$.

$$\mathbb{Q} \text{ is dense in } \mathbb{R} \Rightarrow \exists M_i = \sup_{x \in [x_i, x_{i+1}]} f(x) = 1$$

$$\exists m_i = \inf_{x \in [x_i, x_{i+1}]} f(x) = 0.$$

$$\Rightarrow U(P, f) = 1 \text{ and } L(P, f) = 0.$$

A. Let $f: \mathcal{D} \rightarrow \mathbb{R}$ be a $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable nonnegative function.

Define for $n \geq 1$, and $x \in \mathcal{D}$ $f_n(x) = \begin{cases} \frac{(i-1)}{2^n} & \text{if } \frac{(i-1)}{2^n} \leq f(x) < \frac{i}{2^n} \text{ for } i=1, 2, \dots, n \\ n & \text{if } f(x) \geq n \end{cases}$

prove (i) For each $n \geq 1$, $f_n(x)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

(ii) For each $x \in \mathcal{D}$, $f_n(x)$ is increasing with n (i.e. $\{f_n\}$ is an $\uparrow$ seq. of functions)

(iii) Prove that $f_n(x) \rightarrow f(x)$ pointwise.

$$(i) \quad n=1 \quad f_1(x) = \begin{cases} \frac{(i-1)}{2} & \text{if } f(x) \in [\frac{(i-1)}{2}, \frac{i}{2}) \text{ for } i=1, 2 \\ 1 & \text{if } f(x) \geq 1 \end{cases}$$

$$= \begin{cases} \frac{(i-1)}{2} & \text{if } x \in f^{-1}([\frac{(i-1)}{2}, \frac{i}{2})) \text{ for } i=1, 2 \\ 1 & \text{if } x \in f^{-1}([1, \infty)) \end{cases}$$

$$= \begin{cases} 0 & x \in f^{-1}([0, \frac{1}{2})) \\ \frac{1}{2} & x \in f^{-1}([\frac{1}{2}, 1)) \\ 1 & x \geq 1 \end{cases}$$

$$n=2 \quad f_2(x) = \begin{cases} \frac{(i-1)}{2^2} & \text{if } x \in f^{-1}([\frac{(i-1)}{2^2}, \frac{i}{2^2})) \text{ for } i=1, 2, 3, \dots, 2 \times 2^2 \\ 1 & \text{if } x \in f^{-1}([1, \infty)) \end{cases}$$

$$= \begin{cases} 0 & x \in f^{-1}([0, \frac{1}{4})) \\ \frac{1}{4} & x \in f^{-1}([\frac{1}{4}, \frac{1}{2})) \\ \frac{1}{2} & x \in f^{-1}([\frac{1}{2}, \frac{3}{4})) \\ \frac{3}{4} & x \in f^{-1}([\frac{3}{4}, 1)) \\ 1 & x \geq 1 \end{cases}$$

$$\Rightarrow f_n(x) = \sum_{i=1}^{n2} \left(\frac{i-1}{2^n} \right) \mathbb{I} \left[x \in f^{-1} \left(\left[\frac{i-1}{2^n}, \frac{i}{2^n} \right) \right) \right] + n \cdot \mathbb{I} \left[x \in f^{-1}([n, \infty)) \right]$$

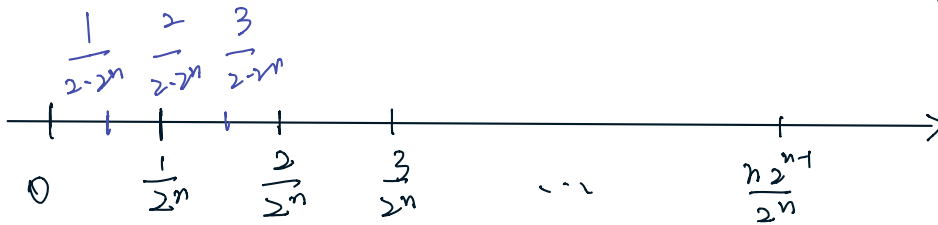
where $I \{x: x \in f^{-1}[\frac{i-1}{2^n}, \frac{i}{2^n})\}$ is $\langle \mathcal{F}, \mathcal{B}_{\mathbb{R}} \rangle$ measurable.

and by corollary 2.1.4. $f_n(x)$ is a simple non-negative $\langle \mathcal{F}, \mathcal{B}_{\mathbb{R}} \rangle$ -mble function.

(ii) $\forall x \in \mathcal{D}$.

$$f_n(x) = \begin{cases} 0 & x \in f^{-1}([0, \frac{1}{2^n})) \\ \frac{1}{2^n} & x \in f^{-1}([\frac{1}{2^n}, \frac{2}{2^n})) \\ \frac{2}{2^n} & x \in f^{-1}([\frac{2}{2^n}, \frac{3}{2^n})) \\ \vdots & \vdots \\ \frac{n \cdot 2^{n-1}}{2^n} & x \in f^{-1}([\frac{n \cdot 2^{n-1}}{2^n}, \frac{n \cdot 2^n}{2^n})) \\ n & x \in f^{-1}([\frac{n \cdot 2^n}{2^n}, +\infty)) \end{cases}$$

$$f_{n+1}(x) = \begin{cases} 0 & x \in f^{-1}([0, \frac{1}{2 \cdot 2^n})) \\ \frac{1}{2 \cdot 2^n} & x \in f^{-1}([\frac{1}{2 \cdot 2^n}, \frac{2}{2 \cdot 2^n})) \\ \frac{2}{2 \cdot 2^n} & x \in f^{-1}([\frac{2}{2 \cdot 2^n}, \frac{3}{2 \cdot 2^n})) \\ \vdots & \vdots \\ \frac{(n+1) \cdot 2^{n-1}}{2 \cdot 2^n} & x \in f^{-1}([\frac{(n+1) \cdot 2^{n-1}}{2 \cdot 2^n}, \frac{(n+1) \cdot 2^n}{2 \cdot 2^n})) \\ n+1 & x \in f^{-1}([\frac{n \cdot 2^n}{2 \cdot 2^n}, +\infty)) \end{cases}$$



For $x \in f^{-1}([0, \frac{1}{2^n}))$

$$f_n(x) = 0, \quad f_{n+1}(x) = \begin{cases} 0 & x \in f^{-1}([0, \frac{1}{2 \cdot 2^n})) \\ \frac{1}{2 \cdot 2^n} & x \in f^{-1}([\frac{1}{2 \cdot 2^n}, \frac{2}{2 \cdot 2^n})) \end{cases}$$

$$\Rightarrow f_{n+1}(x) \geq f_n(x)$$

Similarly, for $x \in f^{-1}([0, \frac{n \cdot 2^n}{2^n}))$

$$\forall x \in f^{-1}([\frac{i-1}{2^n}, \frac{i}{2^n})) \quad f_n(x) = \frac{i-1}{2^n}$$

$$\text{but } f_{n+1}(x) = \begin{cases} \frac{2(i-1)}{2 \cdot 2^n} & x \in f^{-1}([\frac{2(i-1)}{2 \cdot 2^n}, \frac{2i-1}{2 \cdot 2^n})) \\ \frac{2i-1}{2 \cdot 2^n} & x \in f^{-1}([\frac{2i-1}{2 \cdot 2^n}, \frac{2i}{2 \cdot 2^n})) \end{cases}$$

$$\Rightarrow f_{n+1} \geq f_n.$$

$$\text{For } x \in f^{-1}([\frac{n \cdot 2^n}{2^n}, +\infty))$$

$$f_n(x) = n, \quad f_{n+1}(x) \geq n.$$

$$\text{So } f_{n+1} \geq f_n \quad \forall x \in \mathcal{D}.$$

(iii) $\forall x \in \mathbb{R}$.

$$f_n(x) = \begin{cases} \frac{j-1}{2^n} & \text{if } x \in f^{-1} \left(\left[\frac{j-1}{2^n}, \frac{j}{2^n} \right) \right) \\ n & \text{if } x \in f^{-1} \left(\left[\frac{n-2^n}{2^n}, +\infty \right) \right) \quad (f(x) \geq n) \end{cases}$$

For a fixed x . $\exists n_0 \in \mathbb{N}$ s.t. $f(x) \leq n \quad \forall n \geq n_0$.

$$\text{So } f_{n_0}(x) = \frac{j-1}{2^n} \quad \text{if } \frac{j-1}{2^n} \leq f(x) < \frac{j}{2^n}$$

$$\Rightarrow |f(x) - f_n(x)| = \left| f(x) - \frac{j-1}{2^n} \right| = \left| \frac{2^n f(x)}{2^n} - \frac{j-1}{2^n} \right|$$

$$\text{denote } D(x) = x - \lfloor x \rfloor \leq 1 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow |f(x) - f_n(x)| = \left| \frac{D(f(x))}{2^n} \right| \leq \left| \frac{1}{2^n} \right| = \frac{1}{2^n} \rightarrow 0$$

$$\Rightarrow f_{n_0}(x) \rightarrow f(x).$$

B. If $f, g \in L_1(\mathbb{R}, \mathcal{F}, \mu)$

show $\min(f, g) \in L_1(\mathbb{R}, \mathcal{F}, \mu)$ and $\min\left(\int f d\mu, \int g d\mu\right) \geq \int \min(f, g) d\mu$.

$$\textcircled{1} \quad \min(f, g) \leq \min(|f|, |g|) \leq |f| + |g| \in L_1(\mathbb{R}, \mathcal{F}, \mu)$$

$$\begin{aligned} \textcircled{2} \quad & 2 \left[\min\left(\int f d\mu, \int g d\mu\right) - \int \min(f, g) d\mu \right] \\ &= \int (f+g) d\mu - \left| \int (f-g) d\mu \right| - \int (f+g) d\mu + \int |f-g| d\mu \\ &= \int |f-g| d\mu - \left| \int (f-g) d\mu \right| \geq 0 \end{aligned}$$

$$\Rightarrow \min\left(\int f d\mu, \int g d\mu\right) \geq \int \min(f, g) d\mu$$